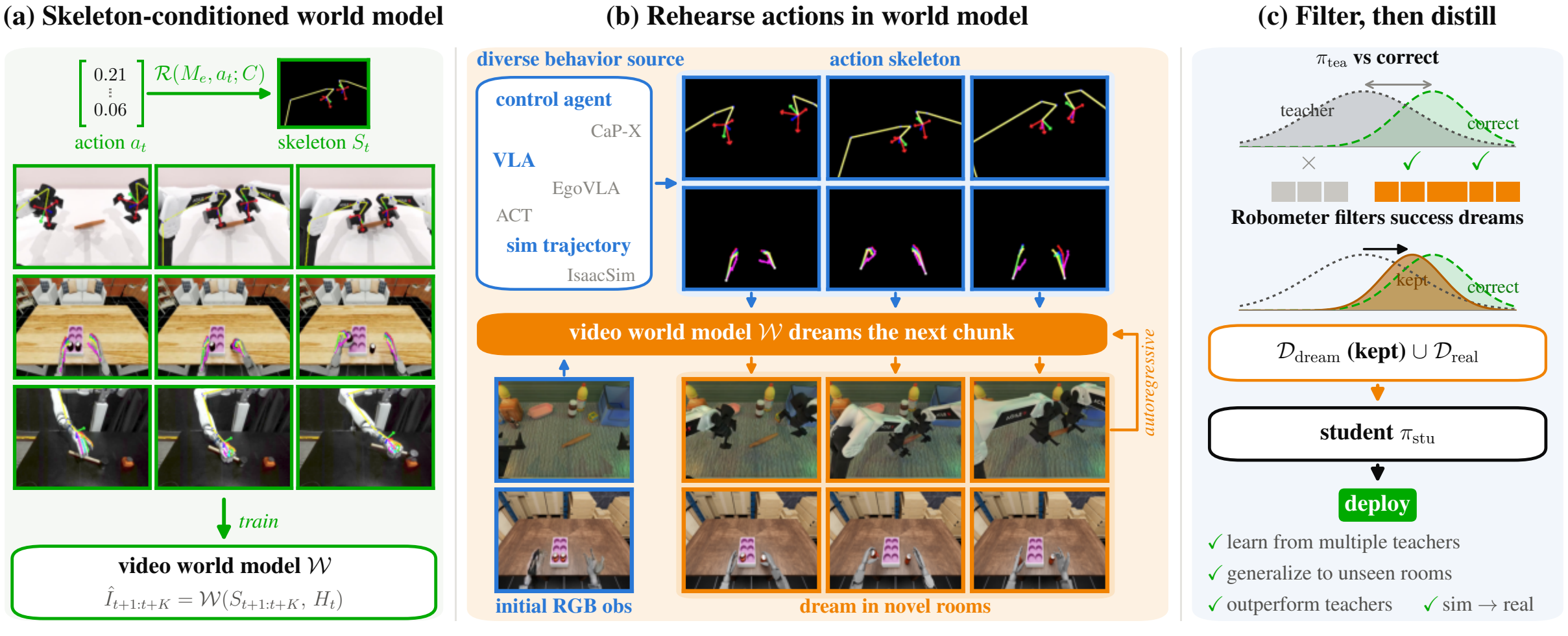

(a) Skeleton-conditioned world model
0.21
0.06
$\mathcal{R}(M_e, a_t; C)$
action a_t
skeleton S_t
train
video world model $\mathcal{W}$
$\hat{I}_{t+1:t+K} = \mathcal{W}(S_{t+1:t+K}, H_t)$
(b) Rehearse actions in world model
diverse behavior source
action skeleton
control agent
CaP-X
VLA
EgoVLA
ACT
sim trajectory
IsaacSim
video world model $\mathcal{W}$ dreams the next chunk
autoregressive
initial RGB obs
dream in novel rooms
(c) Filter, then distill
π_{tea} vs correct
teacher
correct
Robometer filters success dreams
kept
correct
$\mathcal{D}_{\text{dream}}$ (kept) $\cup$ $\mathcal{D}_{\text{real}}$
student π_{stu}
deploy
✓ learn from multiple teachers
✓ generalize to unseen rooms
✓ outperform teachers
✓ sim → real